

Module 04 Data Integration Job Development

IBM InfoSphere Information Server for Data Integration



Module Objectives

After completing this topic, you should be able to:

- Use DataStage Designer
- Understand Stages and Links
- How to Compile and Run Jobs
- Use the Director Client
- Understand the procedure to Import and Export assets
- Create a parameter set and use its parameters in constraints and derivations
- Use the Parallel Job Debugger

DataStage Designer Login

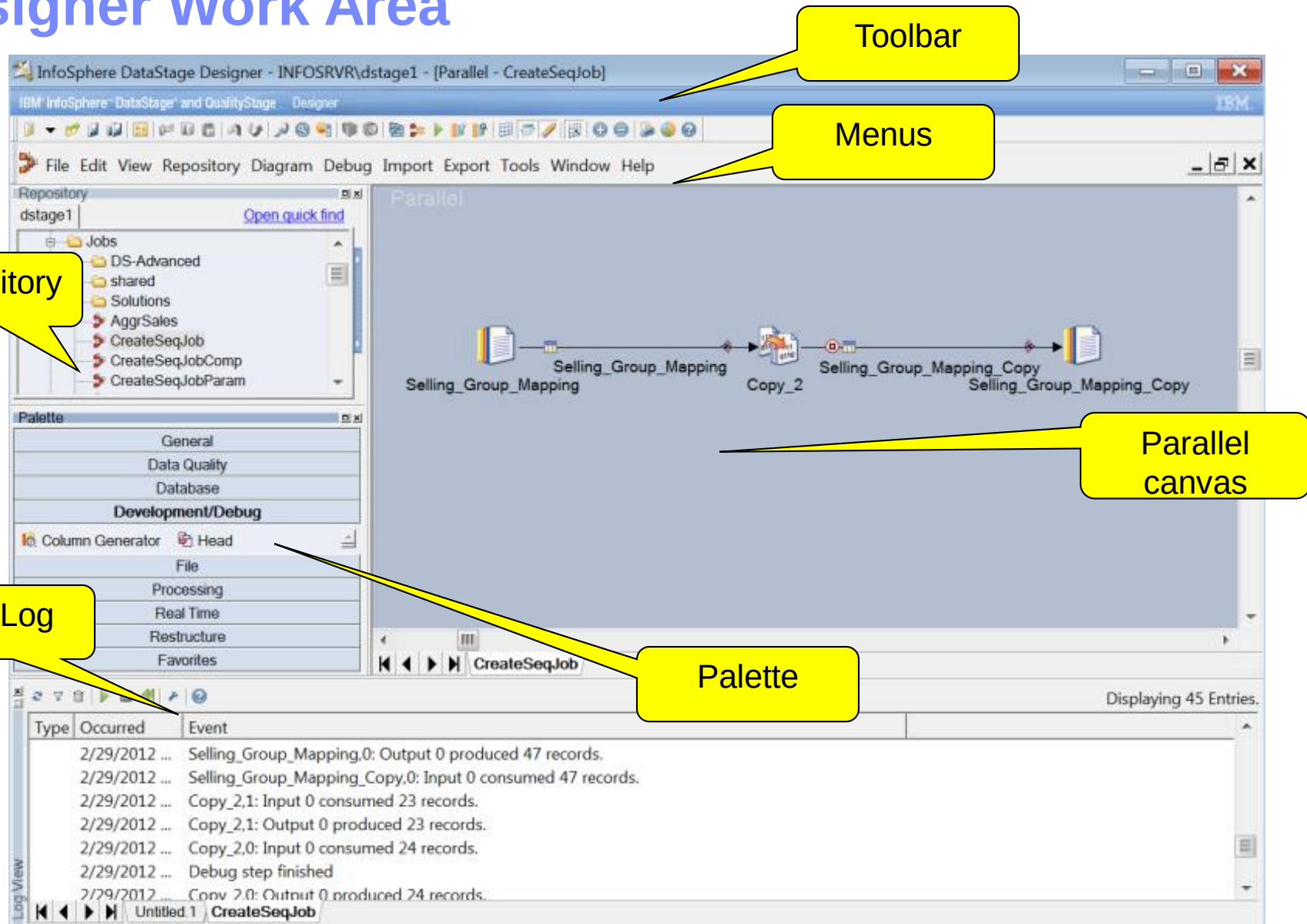
Prior to 11.3 default port is **9080**
After 11.3 already using https with
default port is **9445**.

Host name, port
number of
application server

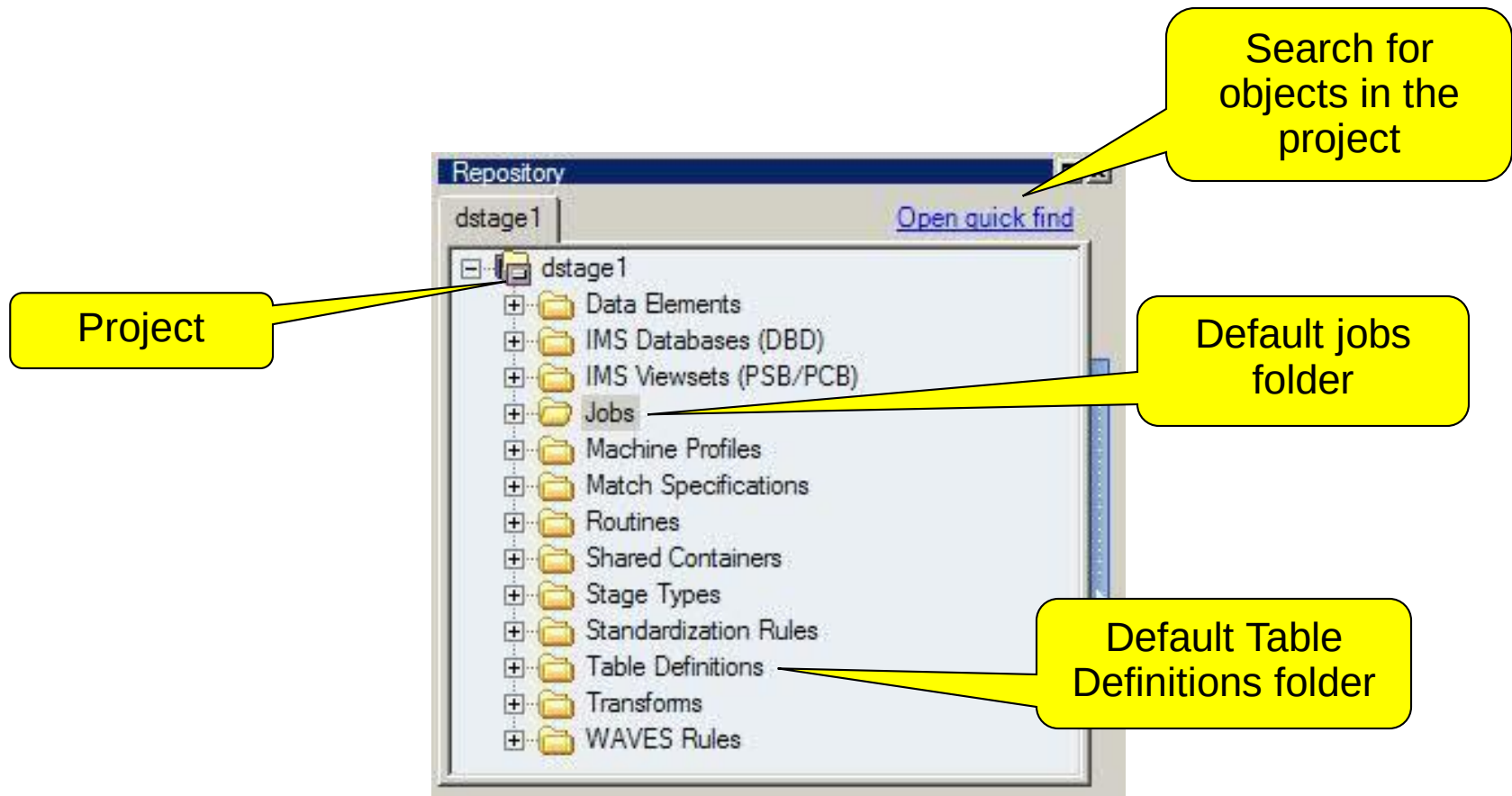
DataStage server
machine / project

The screenshot shows a window titled "Attach to Project" with the IBM logo in the top right corner. The main content area has a blue and white wavy background. The text "IBM® InfoSphere™ DataStage® and QualityStage" is displayed, followed by "Version 11.3". Below this, there are four input fields: "Host name of the services tier:" with a dropdown menu showing "is-server.ibm.com:9445", "User name:" with a text box containing "isadmin", "Password:" with a masked text box showing "*****", and "Project:" with a dropdown menu showing "IS-SERVER.IBM.COM/dstage1". At the bottom, there are two buttons: "Login" and "Cancel".

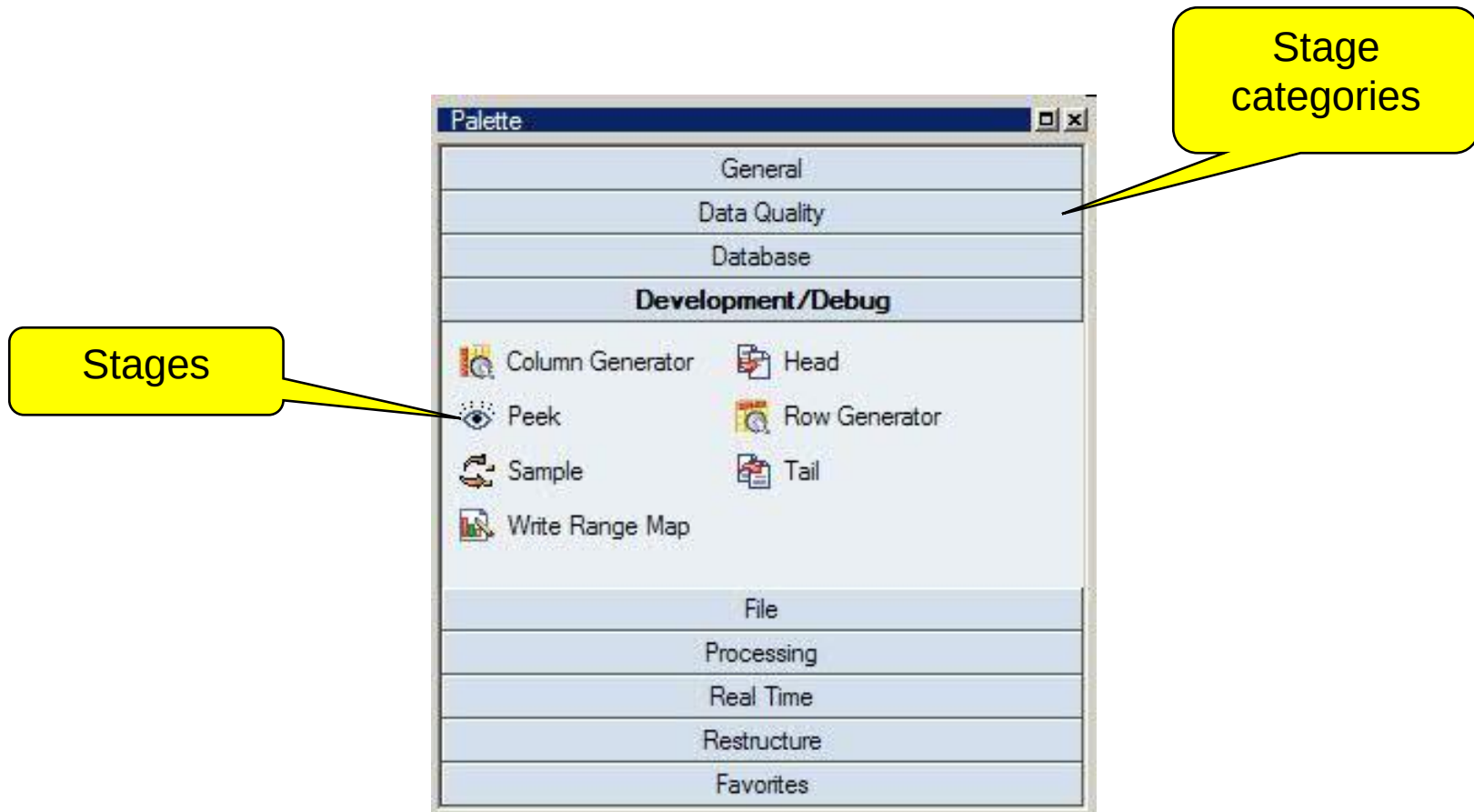
Designer Work Area



Repository Window

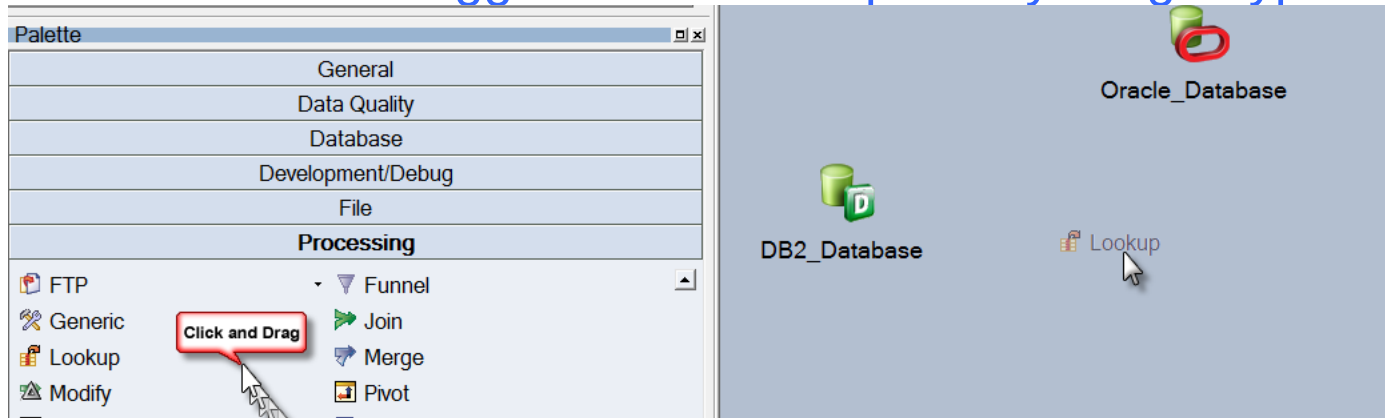


Tools Palette



Adding Stages and Links

- Drag stages from the Tools Palette to the canvas
 - Can also be dragged from the Repository Stage Types folder



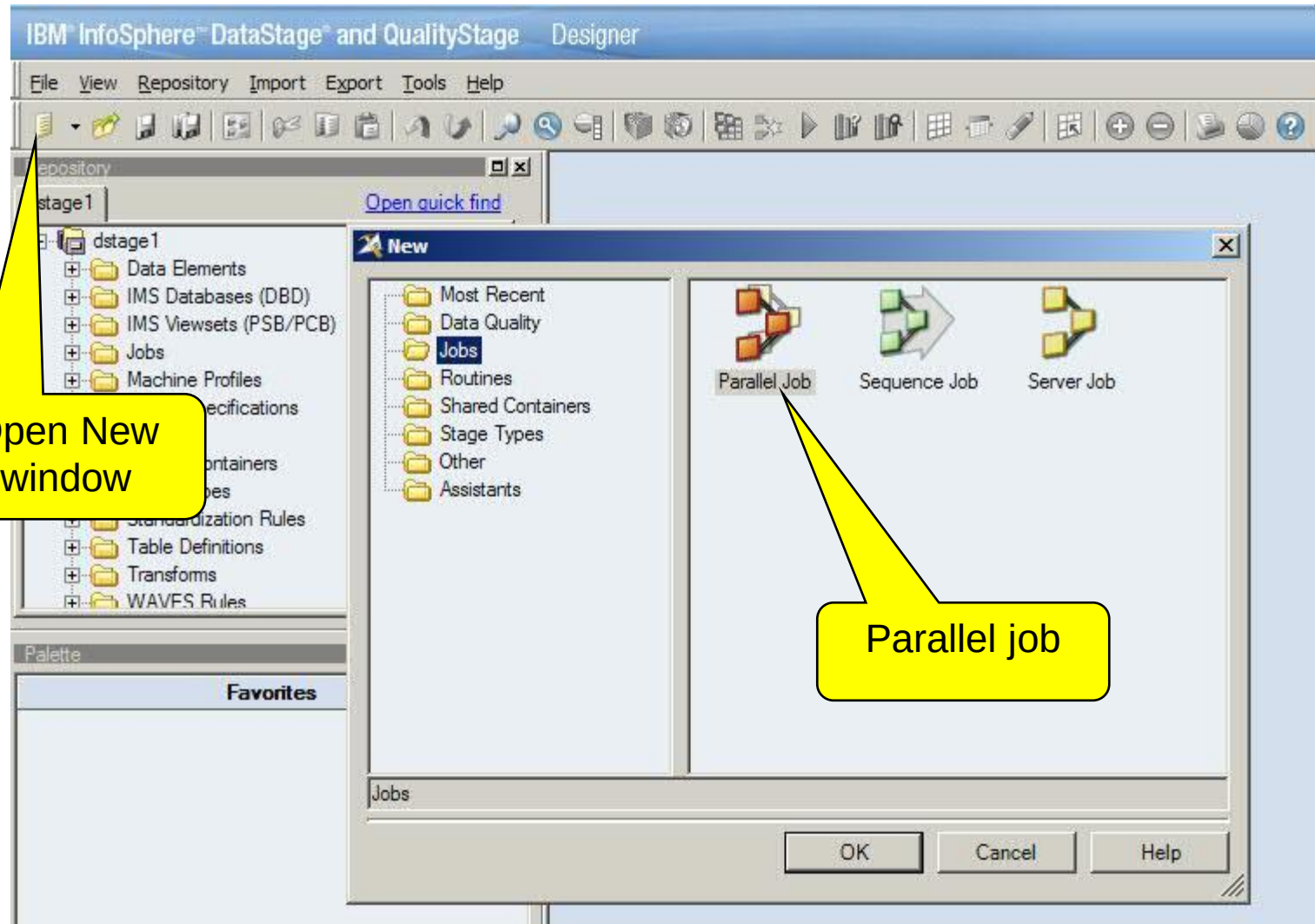
- Draw links from source to target stage
 - Right mouse over source stage
 - Release mouse button over target stage



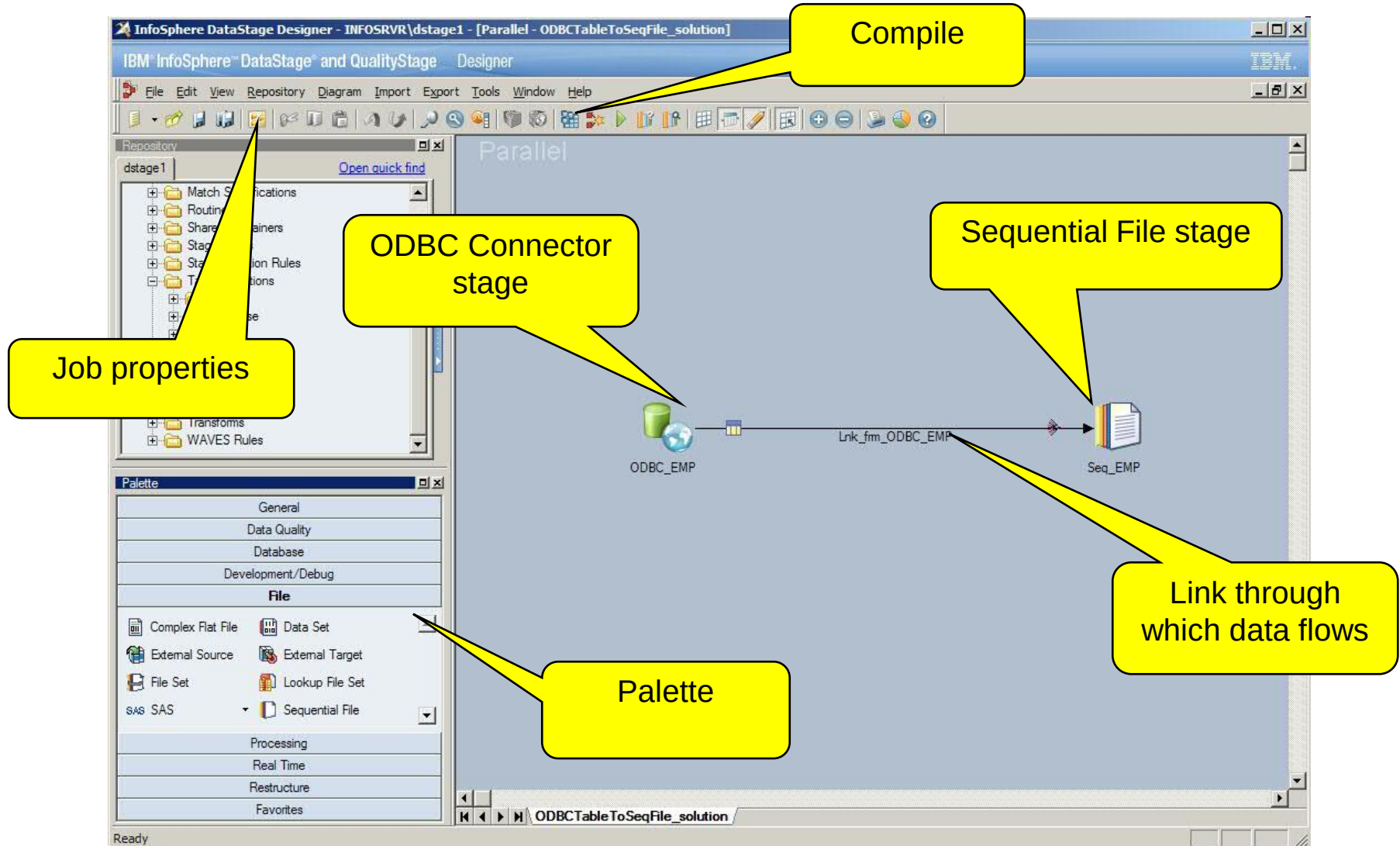
Job Creation Example

- Brief walkthrough of procedure
- Assumes Table Definition of source already exists in the repository
- What the job will do when executed:
 - Reads data from a DB2 table using the ODBC Connector stage
 - Writes data to a sequential file using the Sequential File stage

Create New Parallel Job

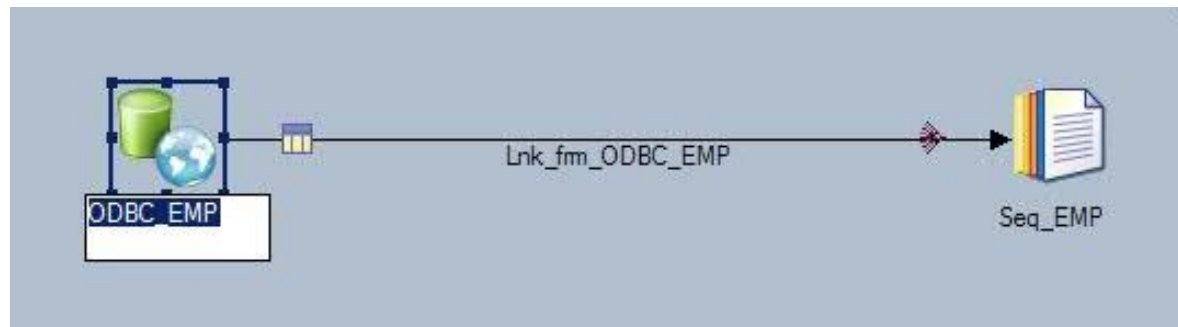


Drag Stages and Links From Palette



Renaming Links and Stages

- Click on a stage or link to rename it
- Meaningful names have many benefits:
 - Documentation
 - Clarity
 - Fewer development errors



ODBC Connector Stage

- Reads data from or writes data to a table in an ODBC data source
 - Reads when it has an output link
 - Writes when it is the target of an input link
- Specify Connector Stage properties
 - Data source name
 - Table name
 - SQL: In our example, we let the stage generate the SQL
- Specify columns
 - Load the table definition on the Columns tab

Inside the Connector Stage

ODBC_EMP - ODBC Connector

Link: Lnk_frm_ODBC_EMP

Type: Output

Target stage: Seq_EMP

Description:

Variant: 3.5

Properties | Columns | Advanced

Connection

Data source *	SAMPLE
Username	db2admin
Password	*****

Usage

Generate SQL	Yes
Table name *	SAMPLE.EMP
Enable quoted identifiers	No
SQL	
Select statement *	
Enable partitioned reads	No
Transaction	
Record count	2000
End of wave	None
Session	
Isolation level	Read uncommitted

OK Cancel Help

Data source

Let the stage
generate the SQL

Table name

Columns Tab

The screenshot shows two overlapping windows from an IBM software interface. The top window is the 'ODBC_EMP - ODBC Connector' dialog, with the 'Columns' tab selected. It displays a table of column definitions. The bottom window is the 'Table Definitions' dialog, showing a tree view of data sources with 'Emp.txt' selected. Yellow callout boxes provide instructions: 'Columns tab' points to the 'Columns' tab in the ODBC Connector window; 'Select Table Definition' points to 'Emp.txt' in the Table Definitions tree; and 'Load a Table Definition' points to the 'Load...' button in the ODBC Connector window.

Columns tab

	Column name	Key	SQL type	Extended	Length	Scale	Nullable	Data element	Description
1	EMPNO	☐	Integer		10		Yes		<none>
2	ENAME	☐	VarChar		10		Yes		<none>
					9		Yes		<none>
					10		Yes		<none>
					10		Yes		<none>
					7	2	Yes		<none>
					7	2	Yes		<none>
					10		Yes		<none>

Select Table Definition

Load a Table Definition

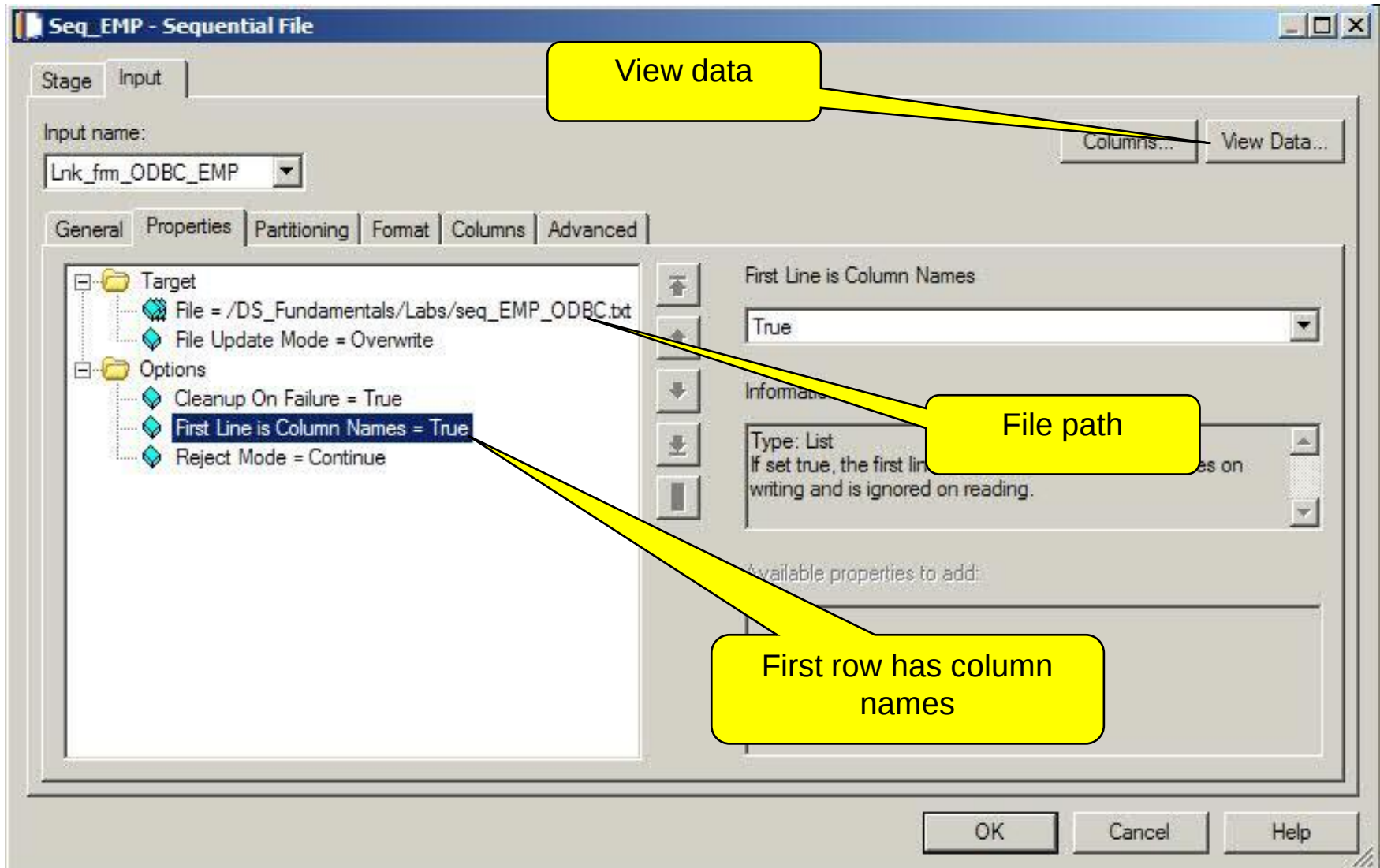
Item name:
Emp.txt

Item type:
Table Definition

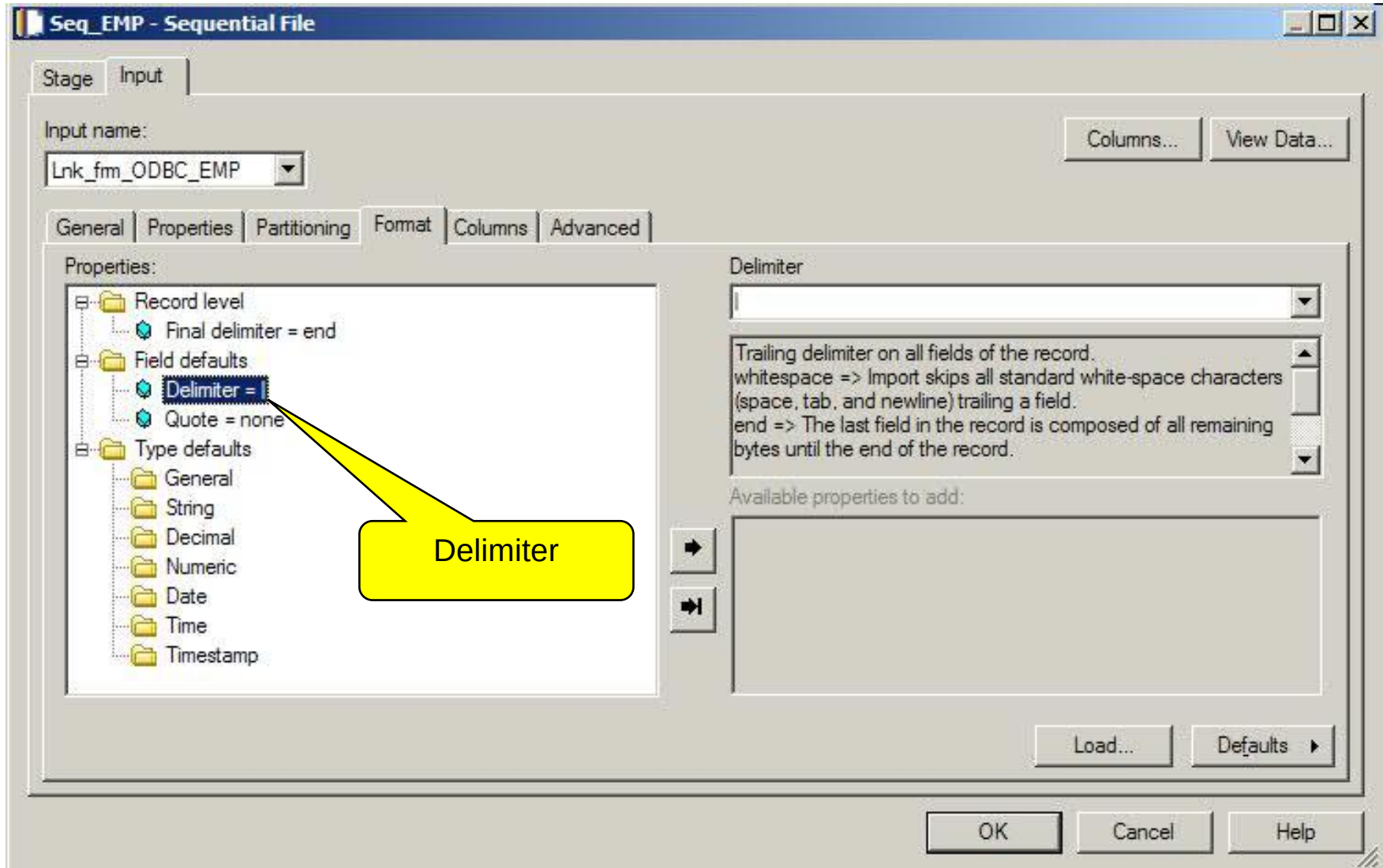
Sequential File Stage

- Reads or writes data from or to a table in sequential file
 - Reads when it has an output link
 - Writes when it is the target of an input link
- Specify Sequential File Stage properties
 - File name and path
- Specify file format
 - Delimiter
- Specify columns
 - Column definitions flow across links, so the columns
 - Defined in the source, Connector stage, will show up in the Sequential File stage

Sequential File Stage Properties



File Format



Defining Columns

Seq_EMP - Sequential File

Stage Input

Input name:
Lnk_fm_ODBC_EMP

Columns... View Data...

General Properties Partitioning Format Columns Advanced

	Column name	Key	SQL type	Extended	Length	Scale	Nullable	Description
1	EMPNO	☐	Integer		10		Yes	<none>
2	ENAME	☐	VarChar		10		Yes	<none>
3	JOB	☐	VarChar		9		Yes	<none>
4	MGR	☐	Integer		10		Yes	<none>
5	HIREDATE	☐	Date		10		Yes	<none>
6	SAL	☐	Decimal		7	2	Yes	<none>
7	COMM	☐	Decimal		7	2	Yes	<none>
8	DEPTNO	☐	Integer		10		Yes	<none>
		☐						

Save... Load...

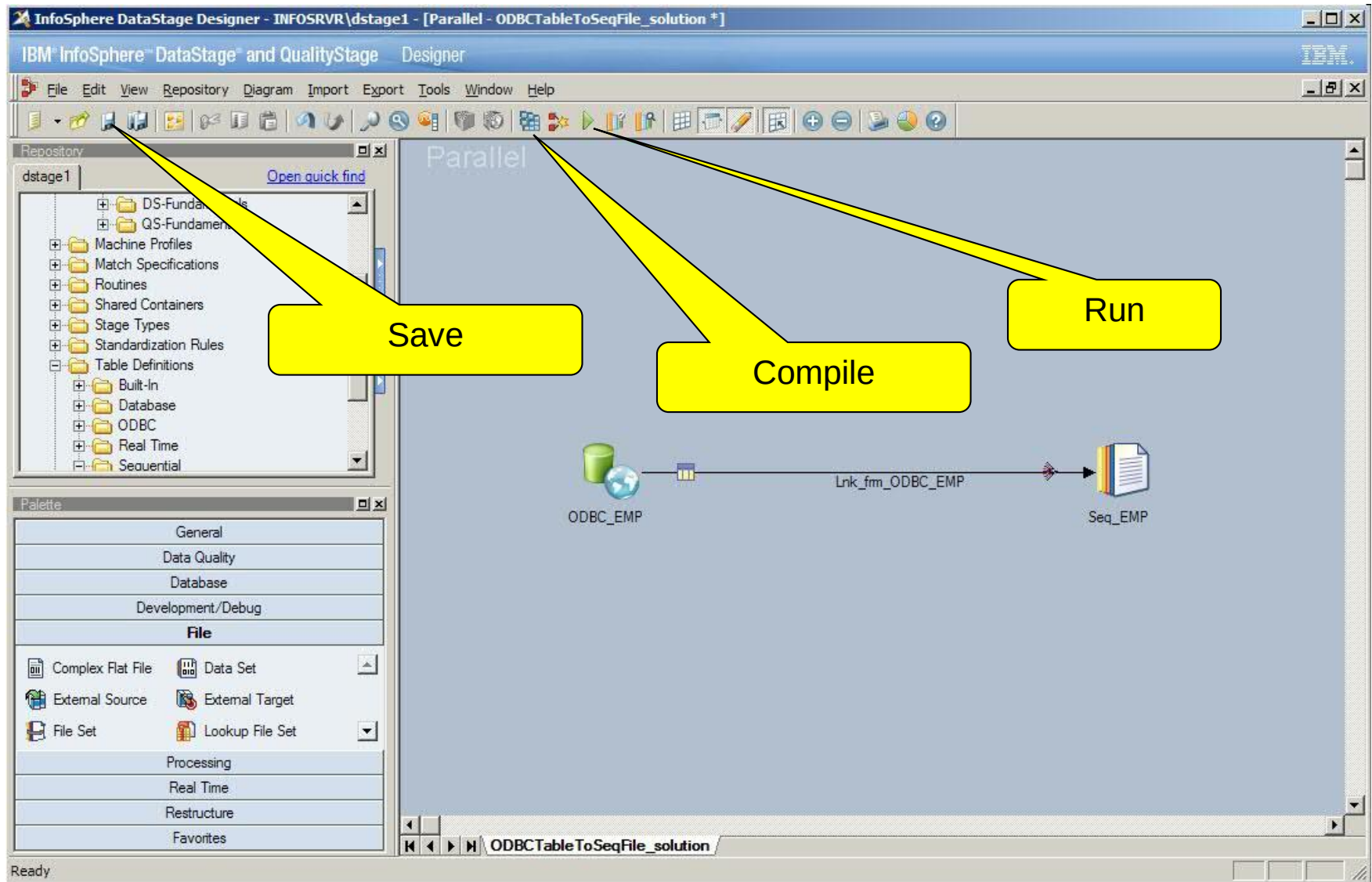
OK Cancel Help

Columns tab

Nullable column

Compiling and Running Jobs

Compiling a Job



Designer Job Log View

- You can view the log of the job run right in your Designer interface
 - View > Job Log

The screenshot displays the IBM Designer Job Log View interface. The top pane shows a data flow diagram with steps: Sales (15 rows, 0 rows/sec), Aggr_Sales, and Revenue (7 rows, 0 rows/sec). The bottom pane shows a log table with columns Type, Occurred, and Event. A yellow callout bubble points to the log entries with the text "Logged messages. Double-click to open the message".

Type	Occurred	Event
	2/28/2012 ...	Sales_Revenue,0: Export complete; 7 records exported successfully, 0 rejected.
	2/28/2012 ...	Aggr_Sales,1: Output 0 produced 1 records.
	2/28/2012 ...	Sales_Revenue,0: Input 0 consumed 7 records.
	2/28/2012 ...	Sales_Revenue.Revenue_Sort,0: Input 0 consumed 3 records.
	2/28/2012 ...	Sales_Revenue.Revenue_Sort,0: Output 0 produced 3 records.
	2/28/2012 ...	Sales_Revenue.Revenue_Sort,1: Input 0 consumed 4 records.
	2/28/2012 ...	Sales_Revenue.Revenue_Sort,1: Output 0 produced 4 records.
	2/28/2012 ...	main_program: Step execution finished with status = OK.
	2/28/2012 ...	main_program: Startup time, 0:00; production run time, 0:00.
	2/28/2012 ...	Parallel job reports successful completion
	2/28/2012 ...	Finished Job AggrSales.
	2/28/2012 ...	Project-level auto-purge, 33 log entries removed

DataStage Director

- Use to run and schedule jobs
- View the job log
- Can invoke directly from Designer
 - [Tools > Run Director](#)

Director Status View

Status view

Log view

The screenshot shows the InfoSphere DataStage Director interface. On the left, a tree view displays the project structure: 'dstage1' containing 'Jobs', 'solutions', 'DS-Fundamentals', 'QS-Fundamentals', and 'Table Definitions'. The 'Jobs' folder is expanded. The main pane shows a table of jobs with columns: Job name, Status, Started, On date, Last ran, and On date. The 'Status' column for all jobs is 'Not compiled'. A yellow callout points to the 'AggrProd_solution' job, with text: 'Select job to view messages in the log view'. The status bar at the bottom indicates 'Status of jobs: 36 entries' and 'Server time: 9/12/2011 03:20 PM'.

Job name	Status	Started	On date	Last ran	On date
AggrProd_solution	Not compiled				
CreateSeqJob_solution	Not compiled				
CreateSeqJobParam_solution	Not compiled				
CreateSeqJobPartition_solution	Not compiled				
CUSTS_solution	Not compiled				
DatasetEMP_solution	Not compiled				
DB2Conn TableToSeqFile_solution	Not compiled				
FilePatternEMP_solution	Not compiled				
FilePatternEMPReaders_solution	Not compiled				
FunnelEmp_solution	Not compiled				
JoinEmpDept_solution	Not compiled				
LKP_Equality_solution	Not compiled				
LKP_Range_solution	Not compiled				
MergeEmpDept_solution	Not compiled				
MultiReadEMP_solution	Not compiled				
NullEmp_solution	Not compiled				
ODBCConn TableToODBCConn Table...	Not compiled				
ODBCTableToSeqFile_solution	Not compiled				
RejectEMP_solution	Not compiled				

Director Log View

Click the notebook icon to view log messages

InfoSphere DataStage Director - INFOSRVR\dstage1

Project View Search Job Tools Help

>Occurred	>On date	Type	Event
3:35:04 PM	9/12/2011	Info	main_program: IBM WebSphere DataStage Enterprise Edition 8.5.0.5946 (...)
3:35:04 PM	9/12/2011	Info	main_program: The open files limit is 1024; raising to 8192.
3:35:04 PM	9/12/2011	Info	main_program: The virtual memory limit is 2922496000 bytes. Raising to 4294967295.
3:35:04 PM	9/12/2011	Info	main_program: conductor uname: -s=Linux; -r=2.6.32.12-0.7-pae; -v=#1 SMP 2010-05-20 11:14:20 +0200; -n=infosrvr; -m=i686
3:35:04 PM	9/12/2011	Info	main_program: orchgeneral: loaded (...)
3:35:06 PM	9/12/2011	Info	main_program: APT configuration file: /opt/IBM/InformationServer/Server/Configurations/default.apr (...)
3:35:06 PM	9/12/2011	Info	Selling_Group_Mapping,0: Import complete; 47 records imported successfully, 0 rejected.
3:35:06 PM	9/12/2011	Info	Selling_Group_Mapping_Copy,0: Export complete; 47 records exported successfully, 0 rejected.
3:35:06 PM	9/12/2011	Info	main_program: Step execution finished with status = OK.
3:35:06 PM	9/12/2011	Info	main_program: Startup time, 0:01; production run time, 0:00.
3:35:06 PM	9/12/2011	Info	Parallel job reports successful completion
3:35:06 PM	9/12/2011	Control	Finished Job CreateSeqJob_solution.
3:41:03 PM	9/12/2011	Control	Starting Job CreateSeqJob_solution.
3:41:16 PM	9/12/2011	Info	Environment variable settings: (...)
3:41:16 PM	9/12/2011	Info	Parallel job initiated
3:41:17 PM	9/12/2011	Info	Parallel job default NLS map UTF-8, default locale OFF
3:41:19 PM	9/12/2011	Info	main_program: IBM WebSphere DataStage Enterprise Edition 8.5.0.5946 (...)
3:41:19 PM	9/12/2011	Info	main_program: The open files limit is 1024; raising to 8192.
3:41:19 PM	9/12/2011	Info	main_program: The virtual memory limit is 2922496000 bytes. Raising to 4294967295.
3:41:20 PM	9/12/2011	Info	main_program: conductor uname: -s=Linux; -r=2.6.32.12-0.7-pae; -v=#1 SMP 2010-05-20 11:14:20 +0200; -n=infosrvr; -m=i686
3:41:20 PM	9/12/2011	Info	main_program: orchgeneral: loaded (...)
3:41:22 PM	9/12/2011	Info	main_program: APT configuration file: /opt/IBM/InformationServer/Server/Configurations/default.apr (...)
3:41:22 PM	9/12/2011	Info	Selling_Group_Mapping,0: Import complete; 47 records imported successfully, 0 rejected.
3:41:22 PM	9/12/2011	Info	Selling_Group_Mapping_Copy,0: Export complete; 47 records exported successfully, 0 rejected.
3:41:22 PM	9/12/2011	Info	main_program: Step execution finished with status = OK.
3:41:23 PM	9/12/2011	Info	main_program: Startup time, 0:03; production run time, 0:00.
3:41:23 PM	9/12/2011	Info	Parallel job reports successful completion
3:41:23 PM	9/12/2011	Control	Finished Job CreateSeqJob_solution.

Log for job: CreateSeqJob_solution 33 entries (filtered) Server time: 9/12/2011 03:43 PM

Logged messages. Double-click to open the message

Import and Export

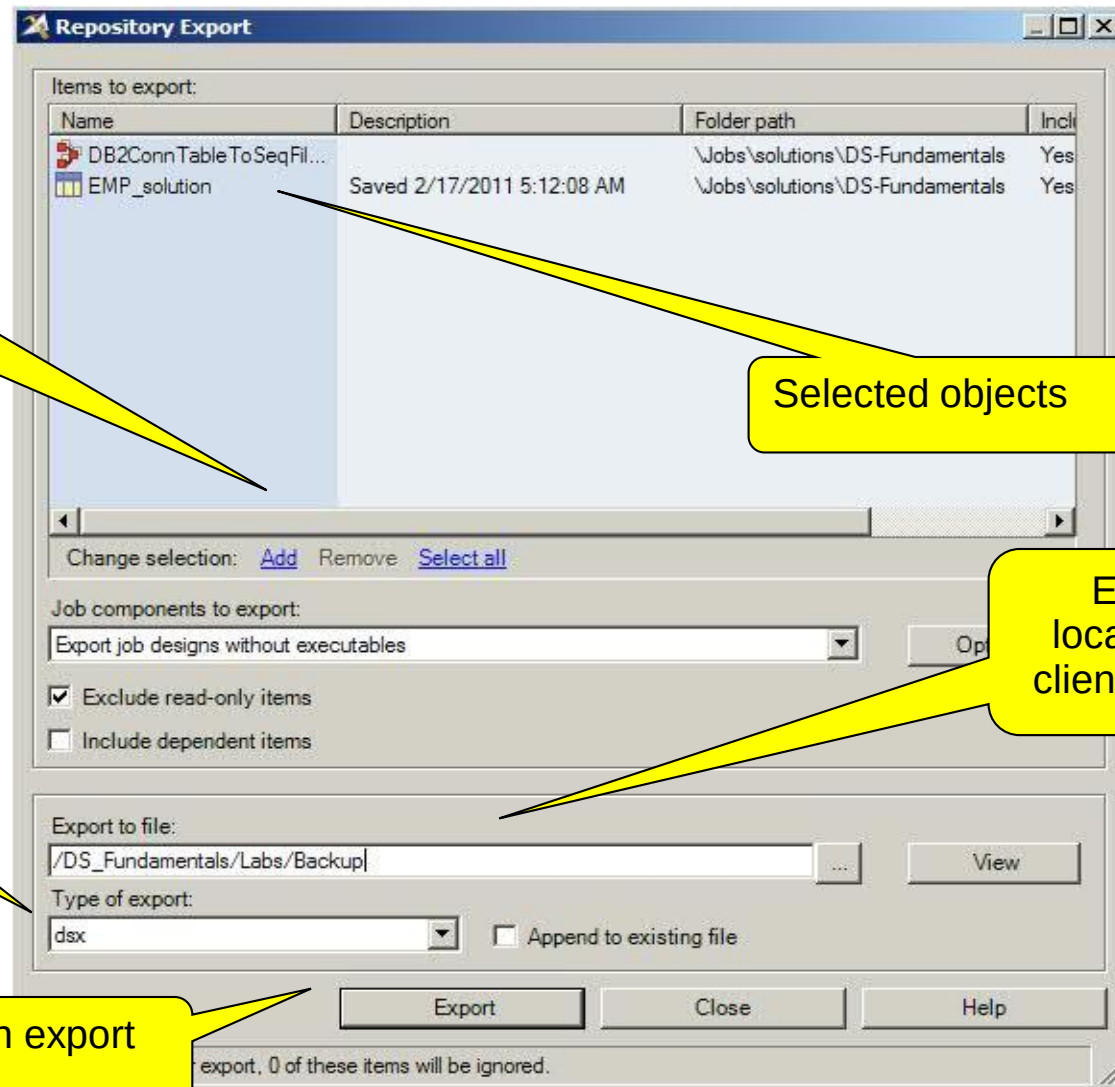
Import and Export

- Any object in the repository window can be exported to a file
- Can export whole projects
- Can be used for backups
- Sometimes used for version control
- Can be used to move DataStage objects from one project to another
- Used to share DataStage jobs and projects with other developers

Export Procedure

- Click “Export > DataStage Components”
- Select DataStage objects for export
- Specify type of export:
 - DSX: Default format
 - XML: Enables processing of export file by XML applications, e.g., for generating reports
- Specify file path on client machine

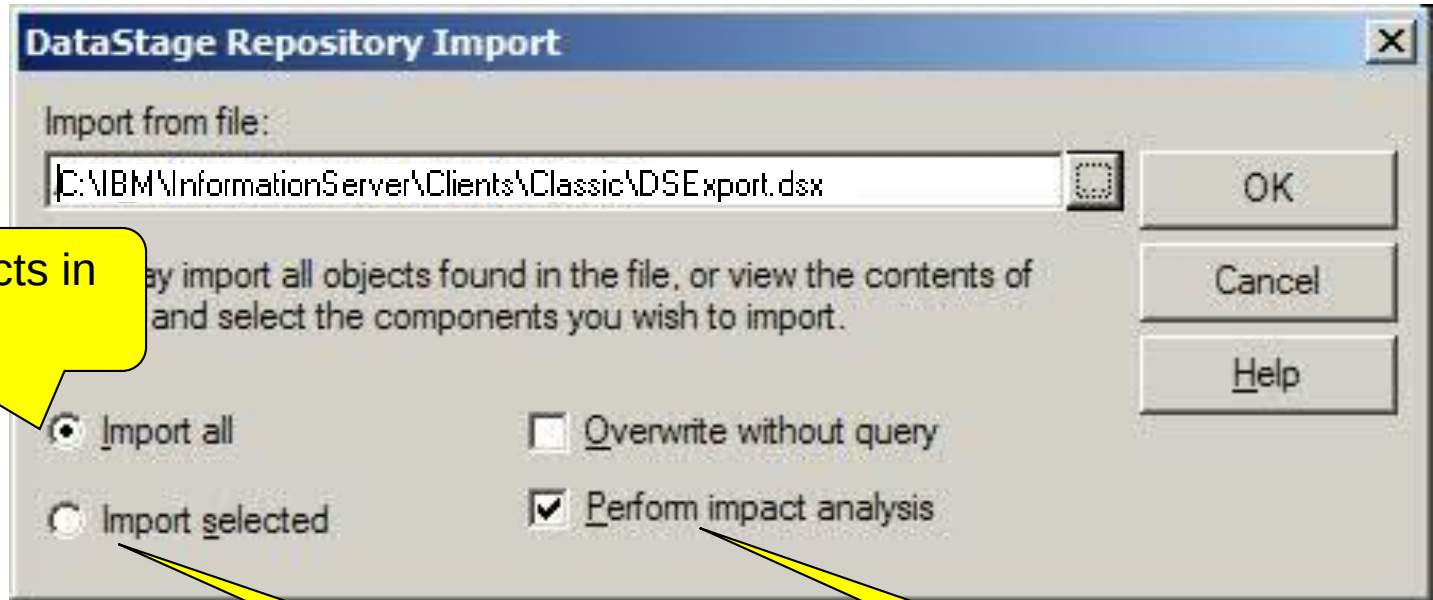
Export Window



Import Procedure

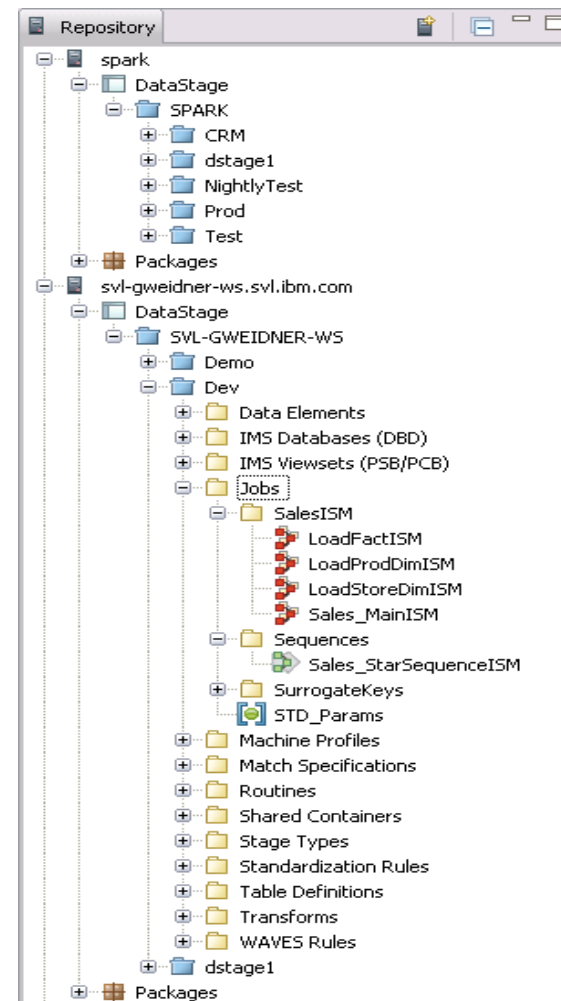
- Click “Import > DataStage Components”
 - Or “Import > DataStage Components (XML)” if you are importing an XML-format export file
- Select DataStage objects for import

Import Options



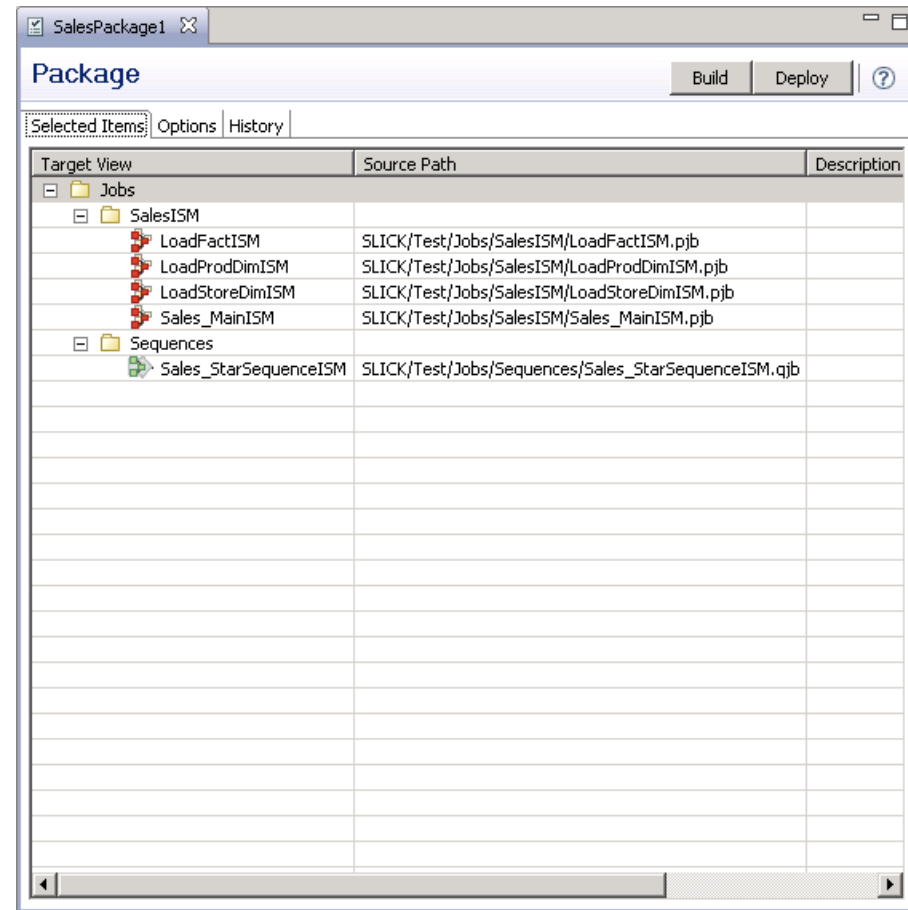
Information Server Manager

- New deployment tool
- Connects to multiple domains
- Shows view of multiple DataStage servers and projects
- Includes corresponding command line interface (istool)
- Provides enhanced export, import and search capabilities



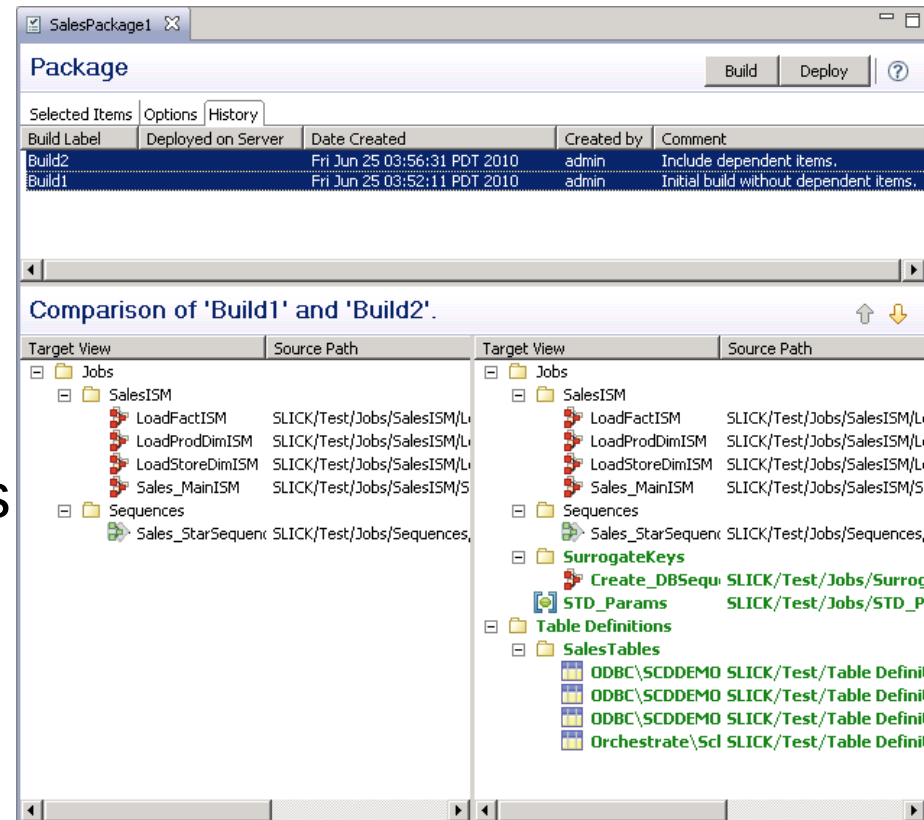
Information Server Manager Deployment Packages

- Package
 - User-defined grouping of related items
- Package file
 - Archive containing design-time and/or run-time data
- Workflow
 - Define
 - Save
 - Build (i.e., create a package file)
 - Deploy



Information Server Manager Deployment Packages

- Build history
 - Previous builds can be maintained and re-deployed
- Build comparison
 - Shows items that were added, deleted or modified
- Only updated items are included in subsequent builds
 - Prevents redundant copies of items in package file



Import/Export Command Line Interface

- Command istool
- Located \IBM\InformationServer\Clients\istools\cli
 - Documentation in same location
- Invoke through Programs > IBM Information Server > Information Server Command Line Interface
- Examples:
 - `istool import –hostname domain:9445 –username dsuser –password inf0server –deployment-package importfile.dsx`
 - `istool export –hostname domain:9445 –username dsuser –password inf0server –job GenDataJob –filename exportfile.dsx`

*Prior to 11.3 default port is **9080**

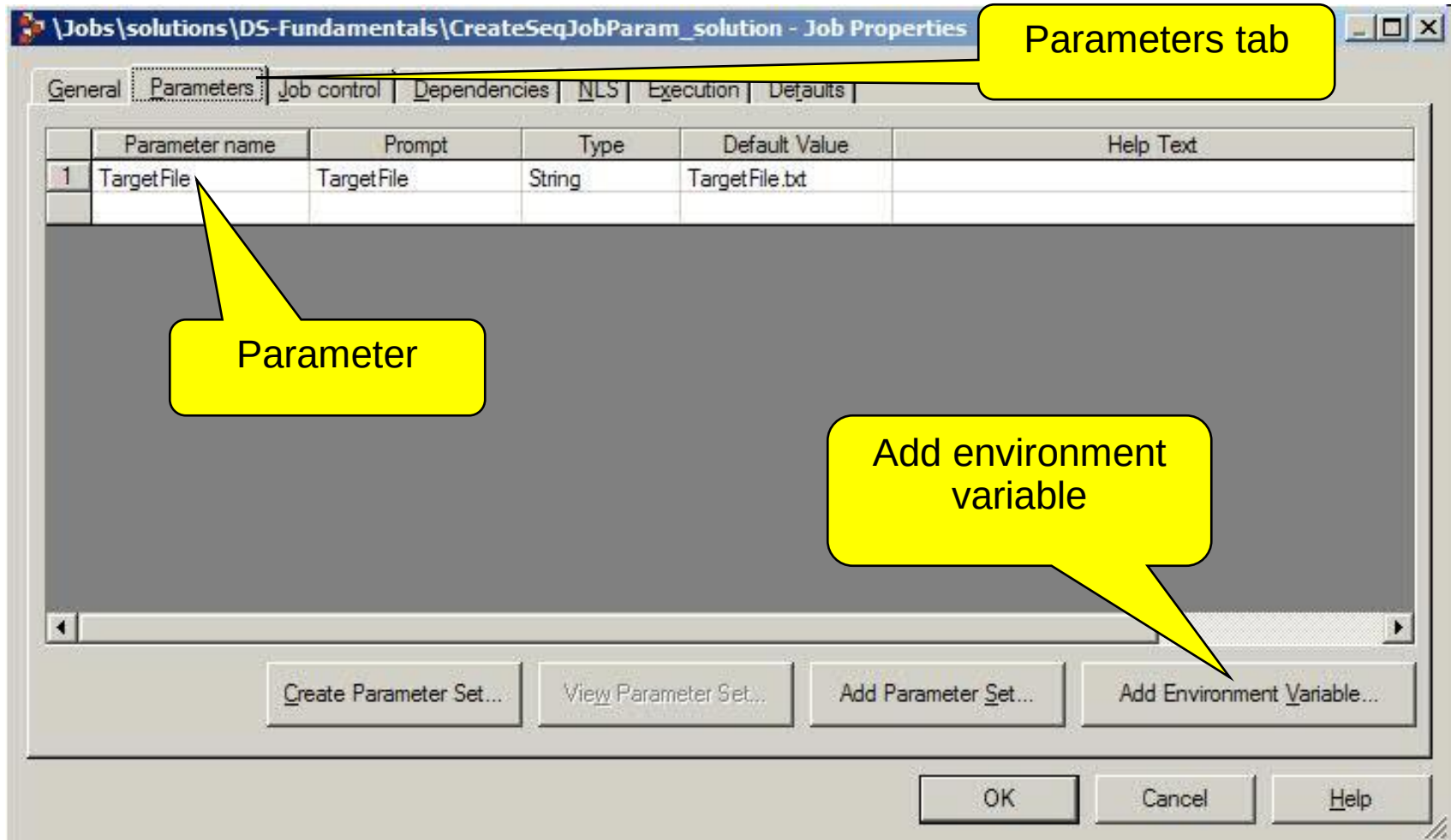
After 11.3 already using https with default port is **9445**.

Job Parameters

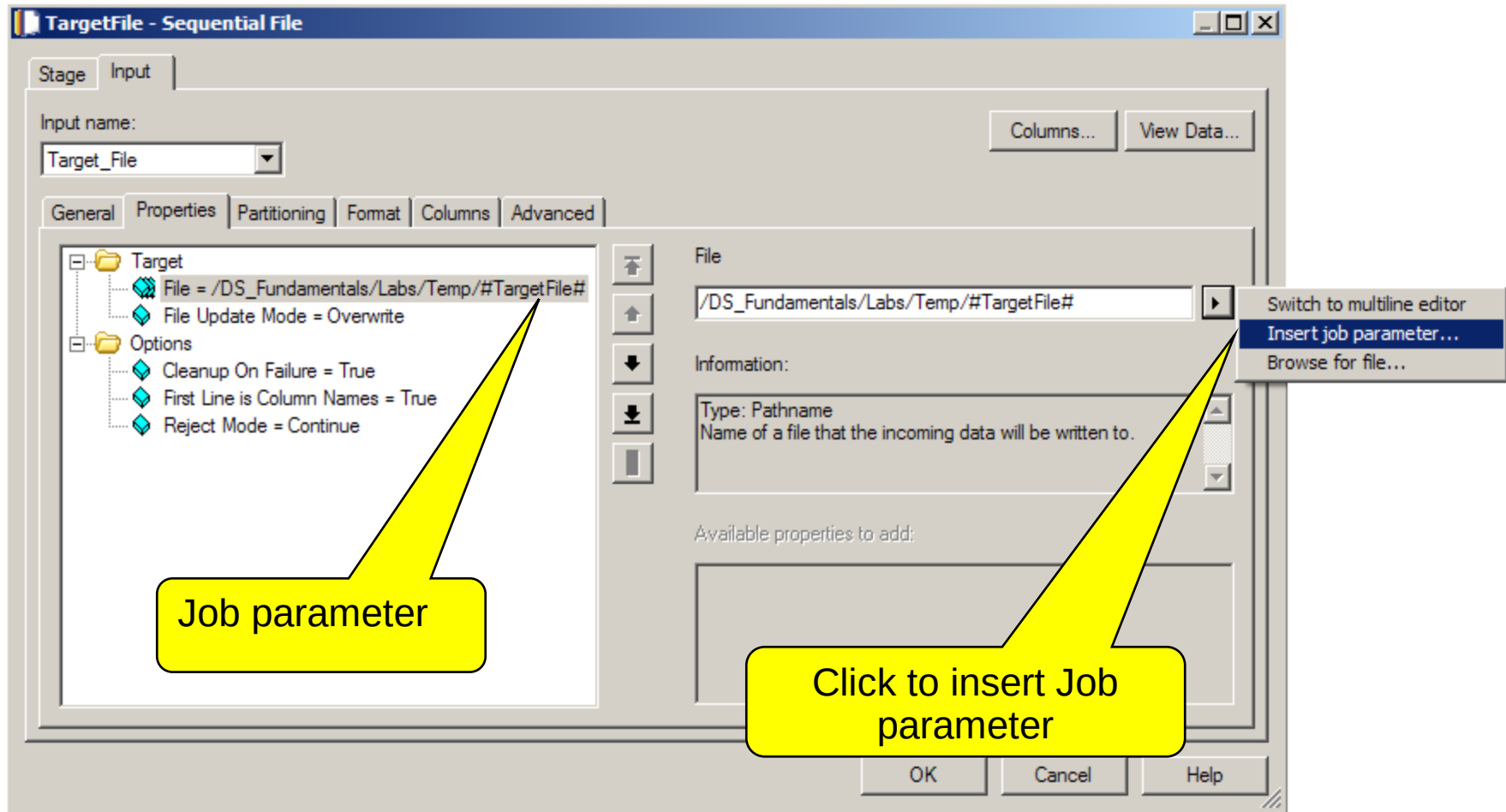
Job Parameters

- Defined in Job Properties window
- Makes the job more flexible
- Parameters can be:
 - Used for directory and file names
 - Used to specify property values
 - Used in constraints and derivations
- Parameter values are determined at run time
- When used for directory and files names and property values, they are surrounded with pound signs (#)
 - e.g., #NumRows#
- Job parameters can reference DataStage environment variables
 - Prefaced by \$, e.g., \$APT_CONFIG_FILE

Defining a Job Parameter



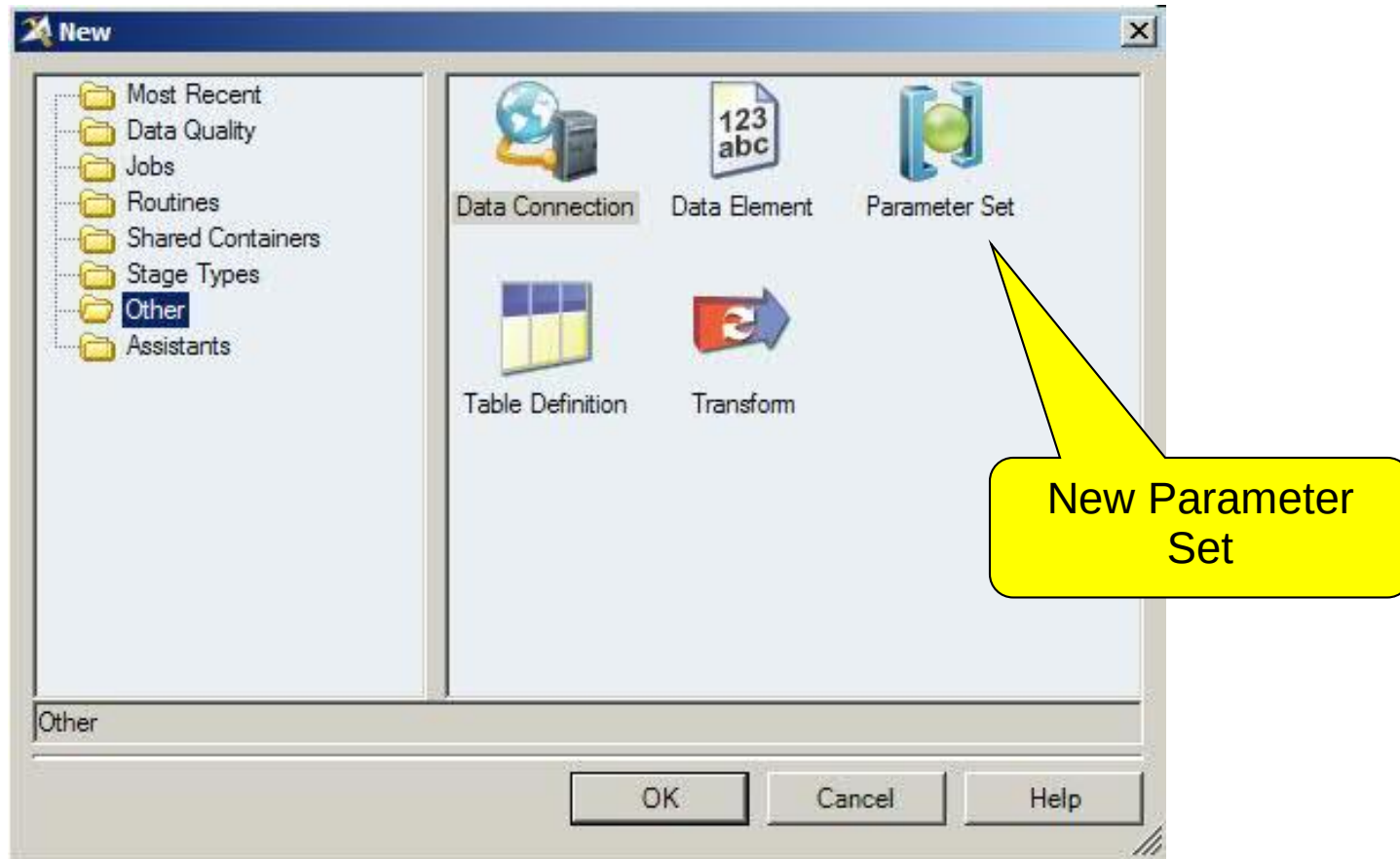
Using a Job Parameter in a Stage



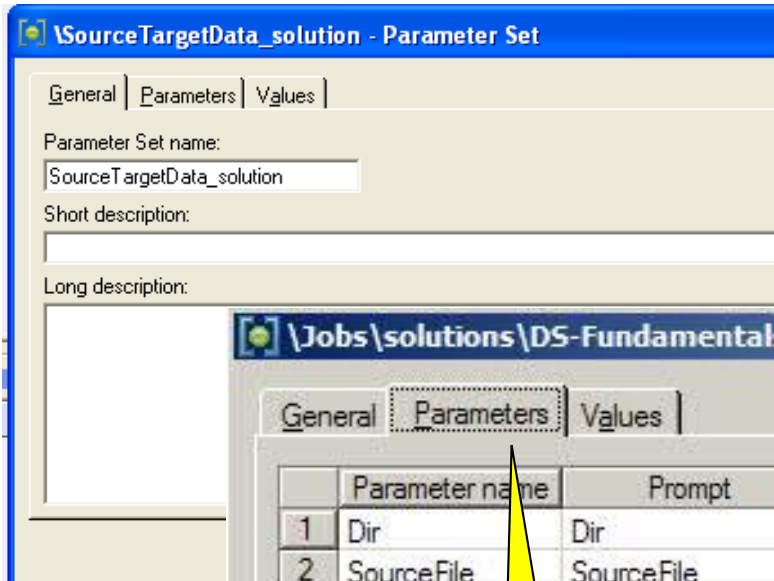
Parameter Sets

- Store a collection of parameters in a named object
- Simplify the creation and management of parameters in the form of name-value pairs
- One or more value file(s) can be specified
 - These files store values for specified parameters
 - The files' values are picked up at runtime
- Parameter Sets can be added to the job parameters

Creating a New Parameter Set



Parameters Tab



SourceTargetData_solution - Parameter Set

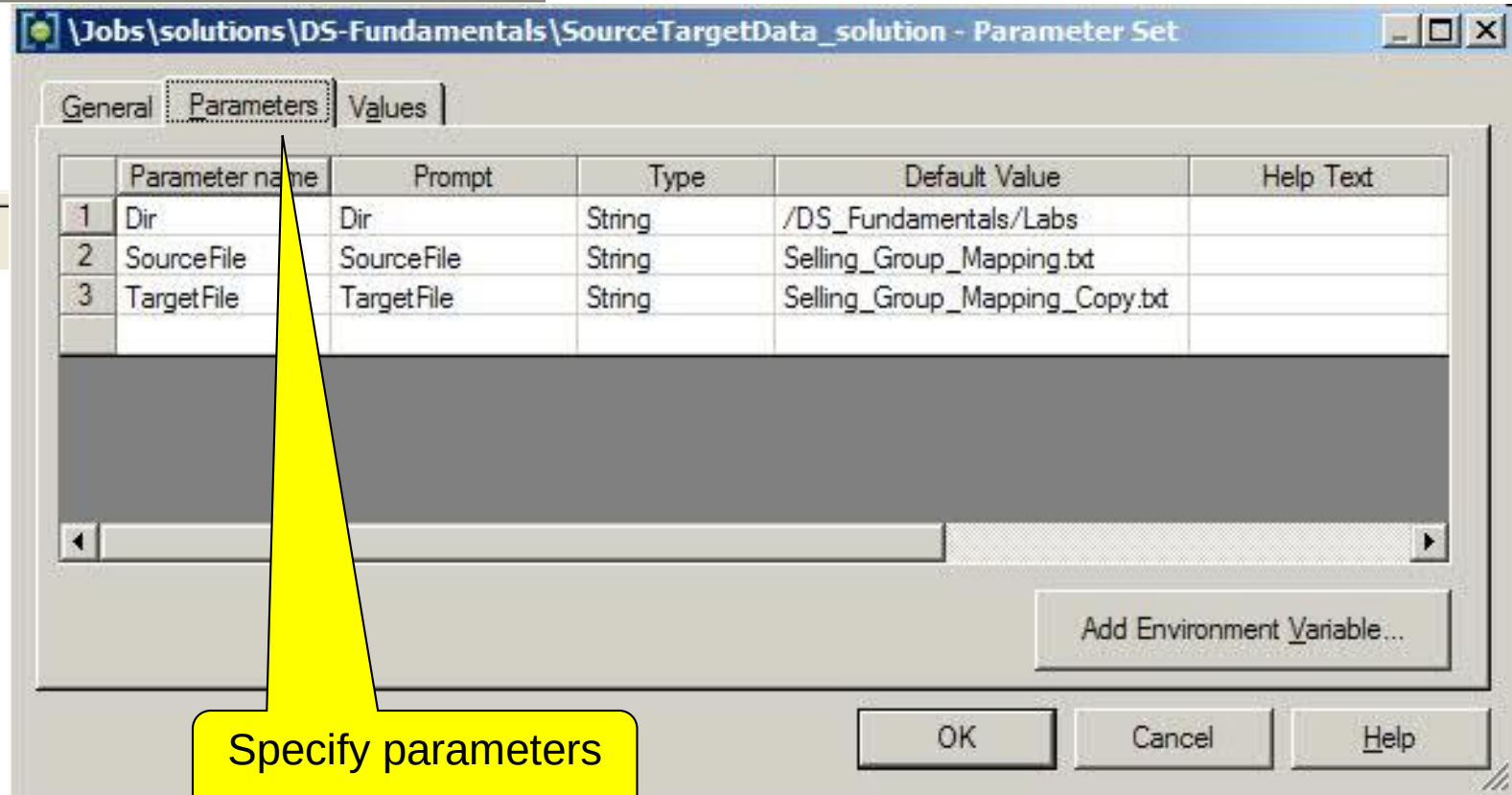
General | Parameters | Values

Parameter Set name:
SourceTargetData_solution

Short description:

Long description:

Parameter set name
is specified on
General tab



Jobs\solutions\DS-Fundamentals\SourceTargetData_solution - Parameter Set

General | Parameters | Values

	Parameter name	Prompt	Type	Default Value	Help Text
1	Dir	Dir	String	/DS_Fundamentals/Labs	
2	SourceFile	SourceFile	String	Selling_Group_Mapping.txt	
3	TargetFile	TargetFile	String	Selling_Group_Mapping_Copy.txt	

Add Environment Variable...

OK Cancel Help

Specify parameters

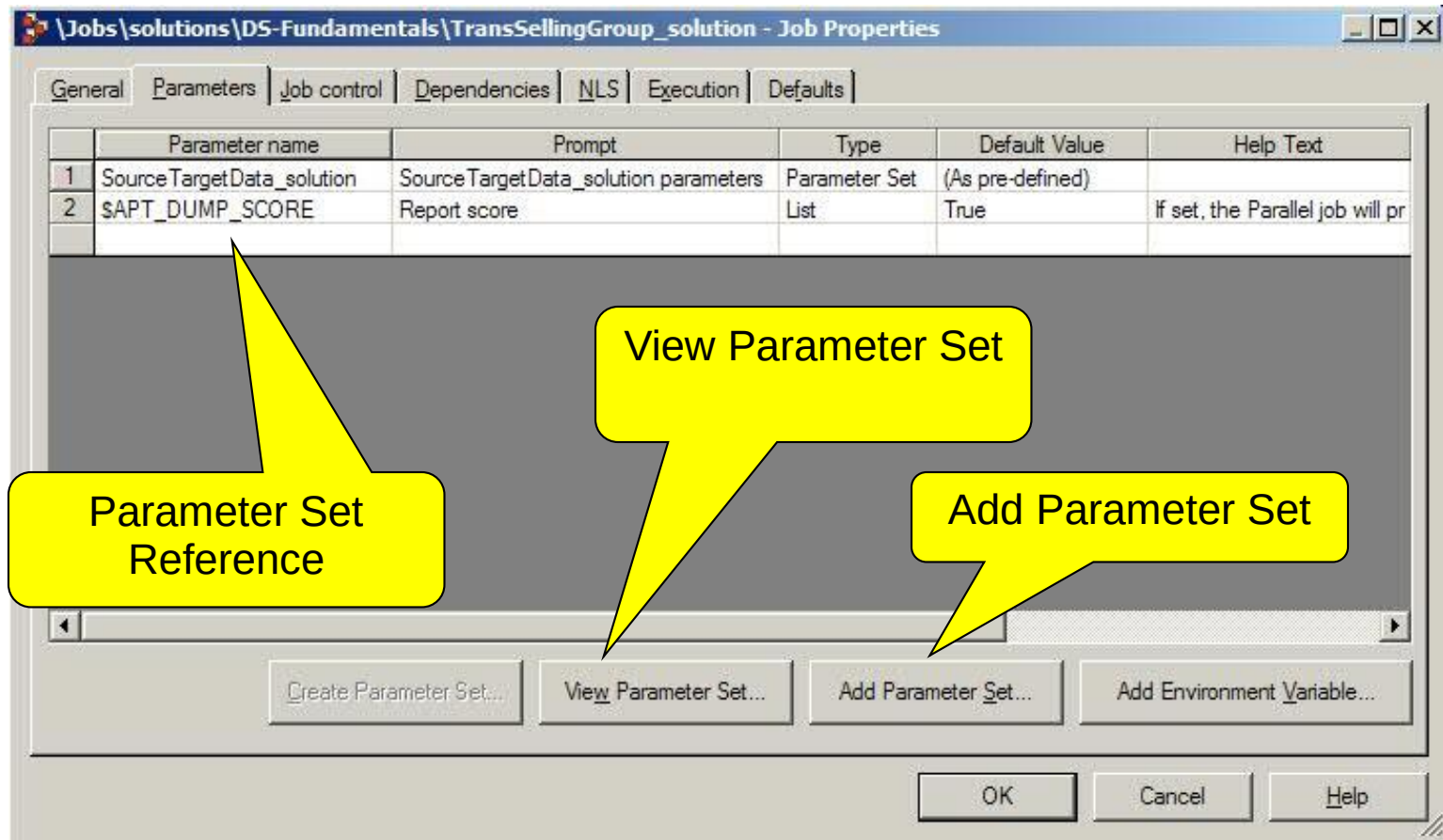
Values Tab

	Value File name	Dir	SourceFile	TargetFile
1	SourceDataValues.txt	/DS_Fundamentals/Labs	Selling_Group_Mapping.txt	Selling_Group_MappingTarget.txt

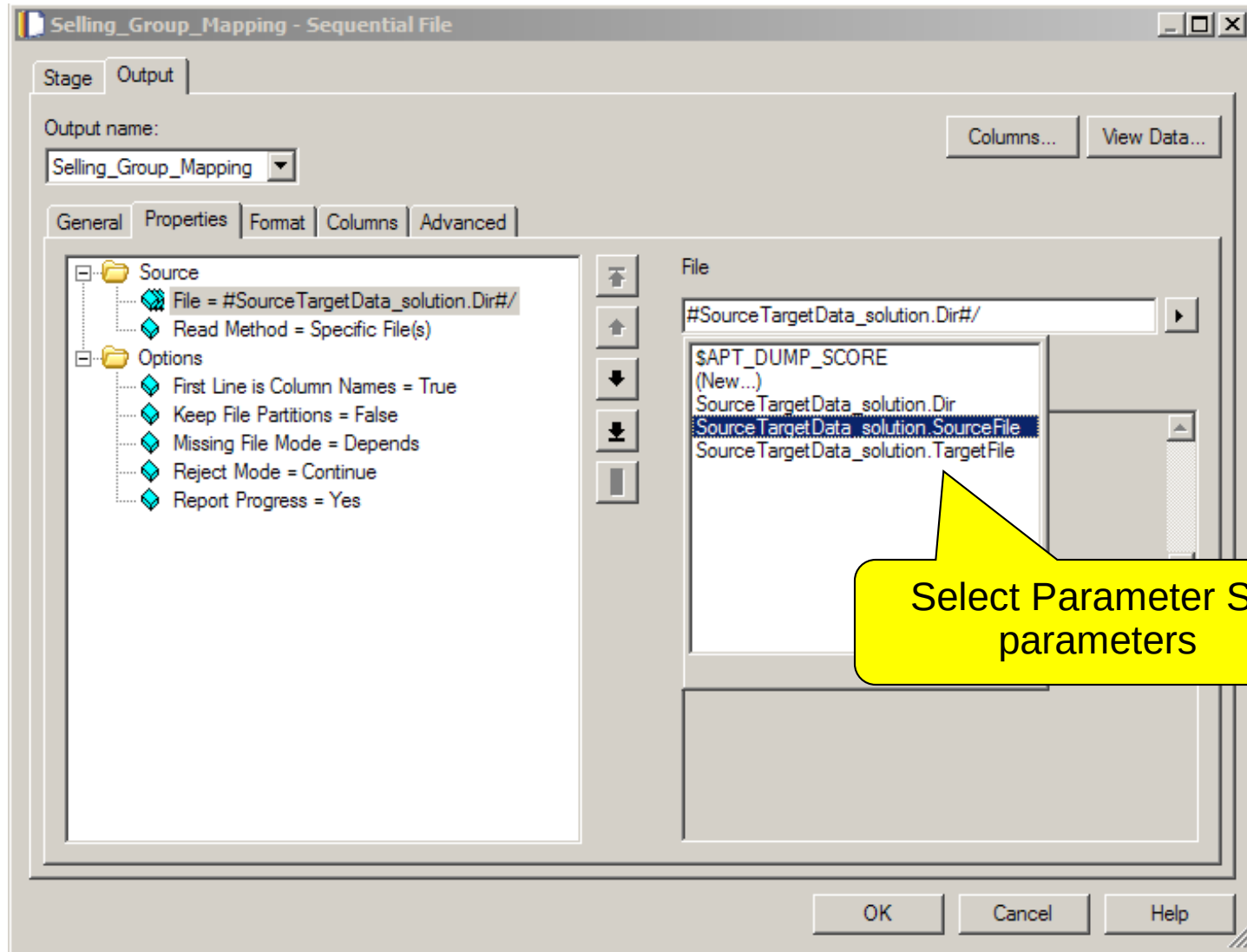
Values file name

Values for
parameters

Adding a Parameter Set to Job Properties



Using Parameter Set Parameters



Debugging Parallel Jobs

Debugging Stages in DataStage

■ Peek Stage

- Prints record field values to the screen.
- Can be used as an intermediate step to peek at the data that is passing through the job or substitute a target stage.



Peek

■ Head, Tail or Sample

- Prints data subsets when dealing with large datasets.



Head



Tail



Sample

■ Row Generator

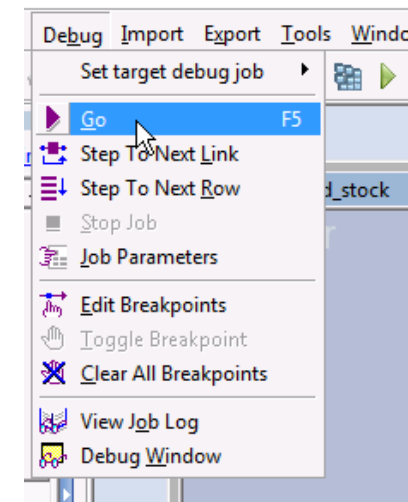
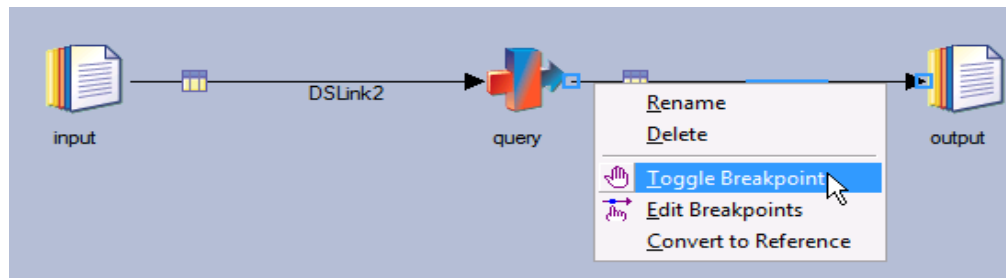
- Produces a set of mock data, in accordance to the table metadata.
- Can be used to test jobs.



Row
Generator

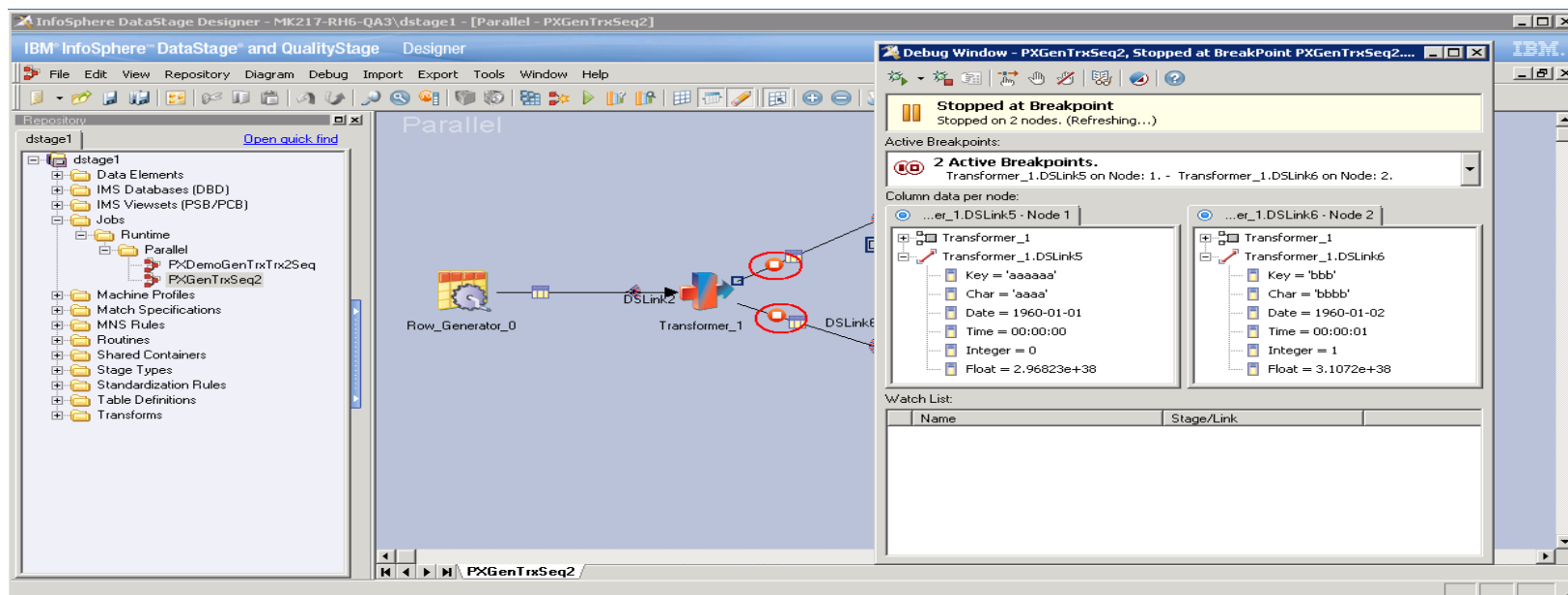
Parallel Job Debugger

- DataStage supports debugging feature for running Parallel Jobs since version 8.7
- Supports multiple breakpoints with conditional logic per link and node
- Visualizes data by node
- Filters complex schemas



Parallel Job Debugger

- From the menu bar, select Debug > Debug Window
- The Debug Window will open While paused at a breakpoint
 - breakpoints can be added/removed
 - row data for the breakpoint link can be examined by node
 - job parameter values can be examined.
 - stage/job property editors can be opened to examine properties of the job design.
 - the running job can be continued or aborted.



Lab 02: DataStage Administration (10')

Lab 03: DataStage Designer (40')